

SPC - IATF 16949 Clause Mapping

Every SPC requirement mapped to IATF 16949:2016 clauses

QUALITY HEAD AGENTS | AIAG SPC 2nd Edition | IATF 16949:2016

Clause 9.1.1 - Statistical Process Control

Requirement:	The organization shall determine, collect, and analyze data to demonstrate the suitability and effectiveness of the QMS.
SPC Requirement:	For special characteristics, SPC is the primary method of demonstrating process control.
Control Charts:	X-bar & R chart for variable data. p-chart or np-chart for attribute data.
Audit Evidence:	Control charts at workstation, updated current shift, all points within UCL/LCL.
Finding Risk:	Control chart not updated, all points on centerline (no real data), or limits set to spec limits = Major NC.

Clause 8.5.1.1 - Control of Special Characteristics

CC Requirement:	Critical Characteristics must be controlled. If $Cpk < 1.67$, 100% inspection is required as interim measure.
Cpk Target:	$Cpk \geq 1.67$ for CC characteristics. If $1.33 \leq Cpk < 1.67$, improvement plan required.
Ppk for PPAP:	PPAP submission requires $Ppk \geq 1.67$ for CC from a minimum 30-piece initial study.
Audit Check:	Auditor will ask for the most recent Cpk for each CC. Must be documented and reviewed.

Clause 8.3.4.4 - PPAP Requirement (Element 9)

PPAP Element 9:	Initial Process Study - $Ppk \geq 1.67$ required for all special characteristics before SOP.
Minimum Study:	30 consecutive pieces from SOP run. Calculate Ppk using STDEV of all readings.
Conditional:	If $Ppk 1.33-1.67$: approved with corrective action plan and timeline to achieve 1.67.
Rejected:	If $Ppk < 1.33$: PPAP rejected. Process must be redesigned before resubmission.

Clause 10.2.3 - 4M Change and SPC

Requirement:	When 4M changes (Man, Machine, Material, Method) occur, process must be re-validated.
SPC Action:	After any 4M change: stop SPC history. Recalculate control limits from new data. Conduct new capability study.
Control Limits:	Using old control limits after a 4M change is a finding. Limits must reflect current process.
Evidence:	4M change log with date of change, new control limit calculation, new Cpk value.